

NAME : MAHKAR KHATANA

Roll No. : 2401010188

ASSIGNMENT - 2 (DBMS)

SEMESTER - IVth

SUBMITTED TO - Mansi Kajal

'Section-A'

- 1) Entity Relationship (ER) diagram
- 2) Relational model
- 3) Object-oriented model
- 4) Integrity constraints

Section - 'B'

Ques 2) Data abstraction means hiding complex database details and showing only necessary information to users.

In a healthcare system, data abstraction helps doctors, nurses and administrators access patient information easily without dealing with complex database structure.

- (a) Simplified interface
- (b) Role-based access
- (c) Improved efficiency
- (d) Data security
- (e) Reduced complexity.

Thus, data abstraction makes patient record management efficient, secure and easy to use.

Date _____
Page _____

Ques (3) Benefits

- (a) Easy to use - Healthcare professionals can access information without technical database knowledge.
- (b) Improved data security - sensitive patient data is protected by restricting access.
- (c) Data consistency - Abstraction ensures that all users view consistent and accurate information.
- (d) Better system maintenance - Database changes can be made internally without affecting users.
- (e) Improved efficiency in healthcare operations - Quick access to (users) patient records helps doctor make faster decisions.

Challenges:

- (a) Implementation complexity
- (b) Performance overhead
- (c) High development cost
- (d) Security risks if poorly implemented
- (e) Training requirements.

Section - 'c'

- Student

(Student-ID (PK), Name, Email, Phone, Department)

- Course

(Course-ID (PK), Name, Credits, Department)

- Attributes

(Proff-ID (PK), Name, Department)

⇒ Relationships

1) Student - Enrolls - Course (m:m)

2) Proff - Teaches - Course (1:m)

